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# Evaluating self-sovereign identity solutions for agricultural supply chains: a systematic review

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Secure identity solutions are crucial to ensure traceability, transparency, and security for the agricultural supply chain. Ensuring a secure, privacy-preserving, and decentralized identity layer is essential for protecting stakeholders and maintaining trust in increasingly complex agricultural ecosystems. This paper provides a systematic review of Self-Sovereign Identity (SSI) solutions through three stages and guided by five research questions (RQs) that guide the assessment and evaluation process. Initially, 1,229 articles were retrieved from multiple scientific databases, and 78 primary studies were selected according to predefined inclusion and exclusion criteria. Although blockchain has been widely studied for traceability, limited work has explicitly evaluated SSI frameworks within agricultural supply chains, revealing a significant gap in identity-focused research. This review assesses the current state of digital identity solutions, with a particular focus on those that utilize blockchain technology for agricultural supply chains. These solutions are evaluated based on the essential principles of self-sovereign identity—transparency, privacy and protection, user control, and decentralization. The findings indicate that while no existing SSI solution fully satisfies all the essential criteria for agricultural supply chains, Sovrin emerges as the most promising framework, excelling particularly in decentralization and privacy protection. The review concludes by identifying key research gaps, outlining deployment-oriented recommendations, and highlighting the need for real-world pilot studies to validate the operational effectiveness of blockchain-based SSI systems in agricultural supply chains.

## KEYWORDS

agricultural supply chain, blockchain, decentralized trust, identity management, privacy, self-sovereign identity (SSI), transparency

## 1 Introduction

The global food system and the diversified rural economies depend on agricultural supply chains that connect producers, processors, distributors, and consumers. These networks are responsible for ensuring that food products get to the right consumers under a variety of circumstances. However, these supply chains face persistent challenges related to traceability, authenticity verification, and data integrity, making them vulnerable to fraud, mislabeling, and opacity in product origin tracking (Menon and Jain, 2024). The lack of transparency and the non-verification of food products' origin can result in a serious compromise of their quality and safety. It can further create a trust deficit among consumers

and stakeholders in these supply chains (Tsoukas et al., 2022; Menon and Jain, 2024). Ensuring the authenticity of producers, certifying agencies, logistics providers, and product attributes is therefore a foundational requirement for modern agri-food ecosystems (Ehsan et al., 2022). The potential of blockchain technology to enhance transparency and security in agricultural supply chains has attracted significant attention (Li K. et al., 2023). However, while blockchain improves transparency, identity verification remains a major challenge, particularly in multi-stakeholder agricultural networks. This challenge is intensified by the fact that agricultural supply chains comprise heterogeneous actors with uneven technological capabilities and varying levels of trust. Blockchain technology offers a variety of options and security features, such as immutability, traceability, and smart contracts, that can provide a high level of transparency and traceability for agricultural supply chains (Ehsan et al., 2022; Dayana et al., 2023). However, these features do not inherently resolve identity verification issues in rural and resource-constrained contexts. The primary concern is how to implement it in rural, resource-constrained areas where most farming occurs, which is the primary source of the supply chain system. In much the same way, technology that enables self-sovereign identity is very promising for creating decentralized identity management systems. Thus, it allows individuals and organizations to control their identity at a global level without relying on national or other central authorities (Čučko and Turkanović, 2021; Bhat et al., 2022). What makes this technology particularly interesting is its potential in contexts where access to essential services is often restricted. Access control and identity verification are crucial for managing the farming supply chain. The centralized model of identity access we currently have is, in many ways, suitable for data privacy and identity management, as it includes intermediaries that provide little control to actual users (Fatima et al., 2022). Despite advances in blockchain and SSI, the literature lacks a unified understanding of how these technologies can work together to address identity, privacy, and traceability challenges in agricultural supply chains.

Assessing the applicability, limitations, and effectiveness of blockchain and self-sovereign identity (SSI) technologies in agricultural supply chains is vital. Although the existing literature focuses on the role of blockchain in promoting transparency, few studies focus on combining these two technologies to secure not only the agricultural supply chain but also all supply chains (Bhat et al., 2022; Baranidharan et al., 2025). However, while blockchain improves traceability, identity verification remains a fundamental and unresolved challenge, particularly in heterogeneous agricultural networks where farmers, cooperatives, distributors, and certification bodies must interact securely without relying on centralized authorities. This gap highlights the importance of Self-Sovereign Identity (SSI), a decentralized identity model that empowers entities to manage, exchange, and verify identity information without intermediaries. SSI offers key benefits—privacy protection, user control, selective disclosure, and cryptographically verifiable credentials—that align closely with the needs of agricultural supply chains.

Despite the increasing research interest in blockchain-based supply chain systems, limited scholarly work has focused specifically on SSI frameworks for the agricultural sector, leaving unanswered questions about their suitability, scalability, and

practical implementation challenges. This systematic review, therefore, examines SSI solutions within agricultural supply chains, evaluating existing frameworks and analyzing their strengths, limitations, and applicability. The review is guided by five research questions that explore challenges in blockchain adoption, available SSI solutions, essential evaluation criteria, comparative performance of SSI systems, and domain-specific strengths and constraints. To ensure analytic rigor, the review follows PRISMA guidelines and systematically synthesizes evidence from 78 primary studies published between 2018 and 2024. A unified framework to assess these two technologies could provide their potential benefits to impact agricultural supply chains and improve transparency and security. Hence, this paper explores the combination of these two technologies. It reviews the principles surrounding SSI and presents a framework for combining the two technologies in the supply chain, especially the agricultural supply chain.

The contributions of this review are as follows:

1. It identifies and categorizes existing SSI frameworks relevant to agricultural supply chains.
2. It develops and applies a structured evaluation based on six essential SSI principles: transparency, privacy, user control, decentralization, scalability, and usability.
3. It critically analyzes the strengths and limitations of leading SSI solutions, such as Sovrin, uPort, Civic, and others.
4. It synthesizes implementation challenges—including regulatory, technical, and operational barriers—and links them directly to findings from the reviewed studies.
5. It outlines actionable recommendations and research opportunities for future SSI deployment in agricultural supply chains.

This article is organized as follows: a comprehensive background on blockchain and self-sovereign identity (SSI) technologies relevant to agricultural supply chains is provided in Section 2. Section 3 details the research methodology, including the data sources, search mechanisms, and inclusion/exclusion criteria used for study selection. The core of the article lies in Section 4, where a comprehensive analysis of the findings is presented, along with the evaluation criteria and SSI principles. Finally, Section 5 concludes the major conclusions, implications, and recommendations for future research.

## 2 Background

This section reviews the literature necessary for conducting this research. It focuses primarily on the essential notions and applications of SSI and blockchain technology in agricultural supply chains. It examines the role of SSI in identity data flows, the applications of blockchain in SSI frameworks, and the potential of these technologies to provide transparency, efficiency, and trust in the agricultural supply chain. The decentralized promise of these twin technologies constitutes an innovative approach to the unique challenges of farming and the agricultural supply chain (Himeur et al., 2025).

## 2.1 Overview of agricultural supply chains

The supply chains that deliver agricultural products to consumers are complex and interdependent. They involve diverse stakeholders, including farmers, processors, transporters, wholesalers, retailers, and consumers. These stakeholders are interconnected by processes that include cultivation, harvesting, processing, and transportation, with each link in the supply chain playing a vital role in ensuring that agricultural products reach market efficiently and safely (Mao et al., 2018; Lubag et al., 2023). The supply chain complexity comes not just from the need to coordinate all these processes, which often unfold across a series of different geographical regions, but also from the tremendous scale and scope of the supply chain system as a whole (Chhetri et al., 2022).

Ensuring transparency across the supply chain is one of the most significant challenges in agriculture and is essential for building trust between stakeholders and consumers (Bhat et al., 2022; Ehsan et al., 2022). Additionally, despite the increasing reliance on agricultural ecosystems—estimated to support 55% of global GDP (Evison et al., 2023)—many systems still lack optimized mechanisms to leverage transparency for improved sustainability and resilience. Stakeholders are found throughout a supply chain; they include everyone involved in the production, processing, and distribution of agricultural and food products. The consumer, who is engaged in consumer testing, surveys, and research, is the final stakeholder who signs off on the value of a supply chain. What is even more remarkable and really unprecedented is that according to a 2023 analysis by PricewaterhouseCoopers (PwC), over half (55%) of the world's gross domestic product (GDP)—equivalent to approximately \$58 trillion—is moderately or highly dependent on nature and its ecosystem services. Despite this substantial reliance, many agricultural supply chains have yet to be optimized to fully leverage these natural contributions for enhanced efficiency, sustainability, and resilience (PwC, 2023).

Another vital issue in agricultural supply chains is traceability. Farming products are perishable, exhibit biological variability, and are subject to a wide range of controls both before and after harvest. For this reason, consistent tracking of the origin and journey of agricultural products through the supply chain is necessary to ensure they meet safety and quality standards (Salah et al., 2019; Shivendra et al., 2021). However, the effort required to manage and process this high volume of data in a near-real-time setting is challenging. A further challenge, not explicitly addressed in traditional systems, is the lack of reliable mechanisms for verifying stakeholder identity, which complicates trust, accountability, and certification processes. Because agricultural supply chains are so long and diverse, with many participants having very different ways of keeping track of things, putting together a product's journey through the supply chain is almost a Herculean task (Shivendra et al., 2021; Bhat et al., 2022). In addition to these traceability and coordination challenges, most agricultural supply chains also lack reliable mechanisms for verifying stakeholder identity, which is increasingly essential for ensuring trust across complex multi-actor networks.

Furthermore, there are no reliable systems for verifying and regulating farming practices. It poses several risks to our environment because unchecked farming can lead to soil erosion and water pollution. Further, it poses serious risks to our health and

the wellbeing of farmers, as unchecked chemical use poses real dangers. Unchecked farm practices can lead to exploitation in several ways, especially by farmers who play by the rules to compete (Hasan et al., 2024). Given all this, what accountability and evidence can agricultural supply chains offer that they are sourcing responsibly? How can they substantiate that they are adhering to fair trade, standards of safety, and quality?

## 2.2 Digital technologies in agricultural supply chains

The swift advancement of digital solutions has enabled new approaches to the agricultural supply chain, delivering innovative means to improve efficiency, enhance traceability, and safeguard data. Supply chain operations in agriculture, just like in any other industry, are increasingly being required to adopt new technologies to remain competitive. It has opened the door to many of the latest technologies for the agricultural supply chain to pursue, including the Internet of Things (IoT), data analytics, and mobile computing. Modern agricultural supply chains are underpinned by the Internet of Things (IoT) (Ibraheem et al., 2022). They provide real-time access to the status of supply chain conditions. Sensors collect data on a variety of situations and levels, such as soil moisture (Karunanayaka et al., 2020; Dhal et al., 2024). And the sensors' accessibility makes it possible to use data to inform decisions efficiently. Consumers of the data make better, faster decisions. And the decisions made because of the use of the sensors tend to be safer and of better quality. You have to pay first to set up the sensors and the monitoring to benefit from this real-time supply chain visibility (Vemuri et al., 2023; Al Mashhadany et al., 2024). These advancements, however, still lack decentralized identity safeguards, prompting the exploration of SSI as a complementary technology to enhance trust and user-controlled authentication.

Another transformative technology for agricultural supply chains is data analytics. It allows stakeholders to gain insights from the ever-increasing volume of data generated by IoT devices and other digital sources. These insights can be vital. They can help supply chain stakeholders predict demand, set prices, and identify potential risks (Raji et al., 2024). They can also help stakeholders understand consumers and market demands more deeply. This understanding makes for better-tuned marketing. It also sometimes makes for better-tuned production decisions with the help of artificial intelligence and machine learning (Sohail et al., 2025). They also help supply chain decision-making in other ways. The vital connections among the stakeholders of the agriculture supply chain are being formed and strengthened in large part through the use of mobile computing (Elufioye et al., 2024; Joel et al., 2024). This technology is helping to bridge the communication gap in remote and rural areas of the world, where many farmers live, and where limited access to traditional telephone and Internet infrastructure poses serious hurdles to information flow (Michels et al., 2020). Mobile applications enable farmers to access this vital supply chain information as easily as by tapping out a text message on a cell phone (Thar et al., 2021). When combined with other emerging digital technologies, mobile computing can deliver even greater productivity and efficiency gains.

In spite of these advances, the Internet of Things (IoT), data analytics, and mobile computing still mainly depend on centralized databases and third-party intermediaries (Omar et al., 2022). This centralized approach is a strategic liability, as it increases current and future systems' vulnerability to data security, manipulation, and transparency issues. Besides, centralized systems often create information asymmetry, the most evident form of which is the excessive control that specific stakeholders (such as system owners) have over data and decision-making. Decentralized solutions, such as blockchain technology and self-sovereign identity (SSI) frameworks, are emerging as potential remedies for these limitations (Islam and Apu, 2024). The decentralized, tamper-proof nature of blockchain's ledger system enables secure recording of transactions and product flow, thereby improving traceability and transparency. By removing intermediaries, blockchain can also reduce costs and enable trust among stakeholders. SSI offers a way to manage identity in a decentralized fashion, allowing stakeholders to control their identity and information. Both of these solutions have the potential to improve privacy and user autonomy (Malik et al., 2021; Oriekhoe O. I. et al., 2024).

## 2.3 Blockchain technology in agriculture

The rapidly emerging technology in the agriculture domain is blockchain, which has become well-known for its potential to improve transparency and traceability in agricultural supply chains. It operates as a decentralized ledger, recording across multiple nodes and in a manner that is both secure and nearly impossible to tamper with (Khalid et al., 2023). When applied to agricultural supply chains, blockchain offers real-time transparency: it enables stakeholders to monitor the entire supply chain with the same level of accessibility as the production team. They can use it to track and verify the quality and safety of agricultural products from the farm through the supply chain (Adewusi et al., 2023; Oriekhoe O. et al., 2024). Here are some of the principal applications for agriculture:

- **Provenance and Traceability:** The blockchain records minute details about the origin and journey of agricultural products, significantly reducing the risks of fraud and counterfeiting (Oriekhoe O. I. et al., 2024).
- **Quality Assurance and Certification:** Blockchain can keep certifications for organic farming, fair trade practices, and food safety standards, providing consumers with verifiable proof of product quality (Gkogkos et al., 2023).
- **Automated Transactions via Smart Contracts:** Blockchain platforms can use smart contracts to automate agreements among parties involved. It can cut down on the costs associated with paperwork and actual transactions, allowing for more streamlined operations among any number of involved stakeholders (Bhardwaj et al., 2021).

Even though the potential applications of blockchain in agriculture are numerous, several challenges hinder large-scale implementation. The means and ways of developing and using blockchain technology can be too costly for small farmers. Even

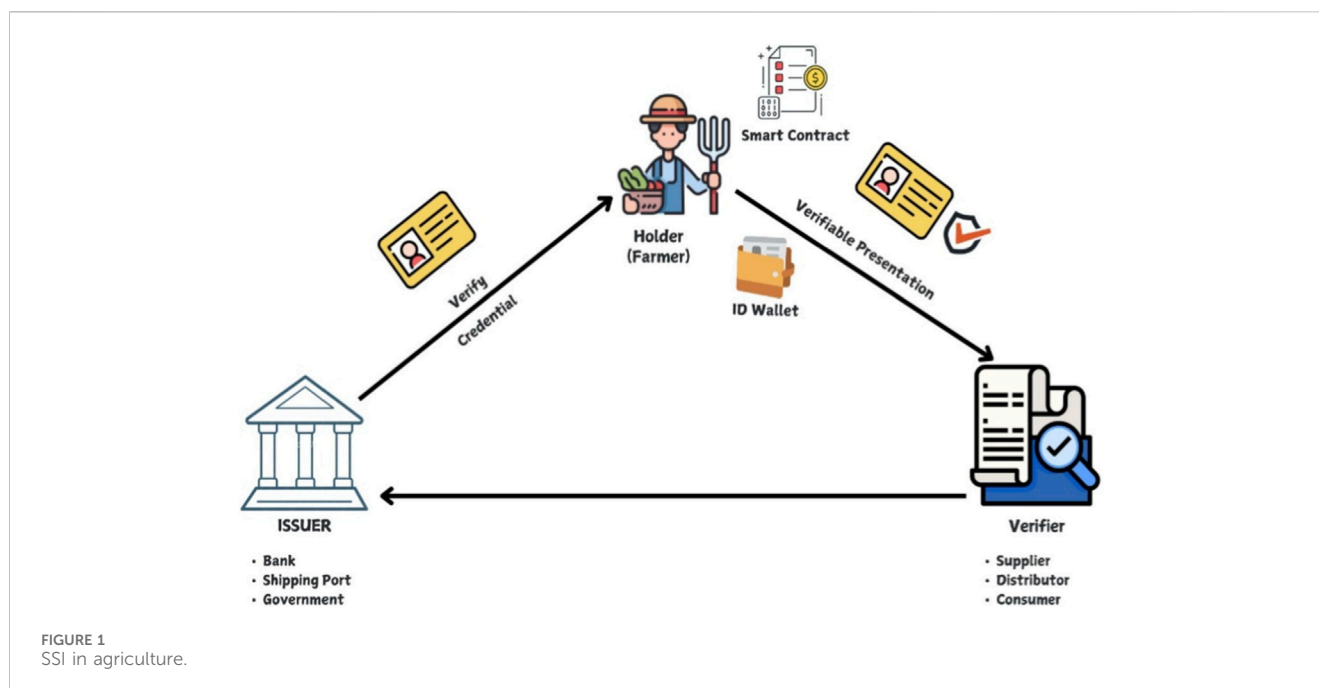
when using less expensive technologies, the issue of network scalability arises. Too many participants and too many transactions can easily and quickly overwhelm a blockchain, which is a major scalability issue (Lin et al., 2020; Xiong et al., 2020). Even more troubling is the fact that some consensus protocols, such as Proof-of-Work, are so energy-intensive that they are sustainable only in regions with abundant energy resources (Lamriji et al., 2023).

In addition, blockchain increases the transparency of systems but does not ensure the privacy of data (Vashishth et al., 2024). Sharing sensitive business information—like pricing strategies or exclusive growing techniques—on a public blockchain for all can leak internal business strategies. Therefore, precise, trustworthy data privacy assurances and clear data ownership frameworks are essential for stakeholders to trust and participate in blockchain-based systems.

## 2.4 SSI in agriculture

Figure 1 shows the basic idea behind Self-Sovereign Identity (SSI) in an agricultural supply chain. It highlights the essential interactions of three main players: the Issuer, the Holder, and the Verifier. In our scenario, trusted organizations—think banks, shipping ports, or government agencies—act as the Issuer. Their job is to create and supply verified credentials to the Holder—things like identity and product authenticity details—and to do so in a way that makes the credentials trustworthy (Wang and De Filippi, 2019; Bhattacharya et al., 2020). For example, a government agency might issue a certified identity for an organic farm to use as part of the farm's verifiable credential set. The organization that is the Holder in our scenario would be the farm itself—a place that serves as a record storage site for a verifiable credential set. The credential set allows the farm to present any number of ID Wallet "Verifiable Credential" instances to any number of "Verifier" instances at any number of times. Verifiable Credentials are tamper-evident digital credentials that can be cryptographically verified and are issued by trusted entities, allowing farmers and other stakeholders to prove specific attributes without revealing unnecessary information.

The ecosystem is further enhanced by smart contracts that automate verification or trigger predefined actions, such as payments, once credentials have been validated (Shuaib et al., 2021). These contracts link the entities, enforcing rules and executing tasks based on the verification of the credential (Gaikwad et al., 2021). The relationships among these entities are critical: the Issuer provides verifiable credentials to the Holder. The Holder shares these credentials with the Verifier. The Verifier checks them against the Issuer or an SSI ledger that uses blockchain technology to ensure they're legitimate (Shuaib et al., 2022a). This system is all about fostering trust among the entities involved. SSI offers a revolutionary approach to digital identity management, allowing individuals and organizations to control their digital credentials without relying on centralized identity providers. In agricultural supply chains, SSI can empower stakeholders—from smallholder farmers to large-scale processors—by enabling them to establish and verify their identities securely. Typically, SSI frameworks use distributed ledger technology to store the verifiable claims that you need to



establish your identity (Stockburger et al., 2021; Ahmed et al., 2022). It ensures that the credentials you use are tamper-proof and, most importantly, trustworthy. Some of the key applications of SSI in agriculture, which range from the basics of identity establishment to more advanced applications like smart contracts, include the following:

- **Farmer Authentication and Certification:** SSI allows farmers to verify their credentials and validate their compliance with organic farming standards. It greatly enhances their access to markets and their credibility in those markets (Cocco et al., 2021; Yildiz et al., 2021).
- **Data Privacy and Selective Disclosure:** SSI permits stakeholders to share only the essential information, thus safeguarding sensitive data like production methods while guaranteeing transparency regarding compliance matters (Schardong and Custódio, 2022).
- **Decentralized Marketplaces:** SSI supports a decentralized mechanism where farmers and consumers can engage directly, with the identity and credibility of their product ensured through verifiable credentials (Belchior et al., 2020).

Although it has many positive aspects, the adoption of SSI in agriculture is constrained by several factors, including non-interoperability, technical complexity, and lack of awareness. Standardized evaluation methods are also needed to assess whether an SSI framework is working well and to validate its compliance with the key principles of identity management that are essential for its widespread adoption.

## 2.5 Integration of blockchain and SSI

The integration of blockchain with self-sovereign identity (SSI) provides a method for dealing with the shortcomings of traditional

digital supply chain systems (Shuaib et al., 2022c). These systems typically count on centralized identity providers, such as governments or companies, that are vulnerable to manipulation and security threats. Blockchain ensures that every participant can be identified, every action can be traced, and every action has been performed in a secure, albeit transparent, manner.

Blockchain-based supply chain systems depend on smart contract functionalities for automated decision-making. Smart contracts have several limitations, such as their dependence on specific platforms and scalability issues.

In this case, the combination with SSI is significant; blockchain cannot foretell all the identity management issues (which supply chains have) and data privacy concerns (which purchasers of supply chain data have) that are also associated with digital centralization. Key applications of blockchain-SSI frameworks in agriculture include:

- **Enhanced Identity Verification and User Control:** With a self-sovereign identity, stakeholders own their digital identity and manage it without relying on centralized authorities.
- **Decentralized Certifications:** SSI allows for independent verification of certifications on a blockchain, ensuring accountability.
- **Privacy-Preserving Data Sharing:** With selective disclosure, SSI enables stakeholders to share only what they need, when they need it, without making all their private information public.

These benefits come with several challenges, such as scalability, interoperability, and technical complexity. Also, there are a few reviews of the integration of blockchain and SSI in the agricultural supply chain. It makes more precise criteria for assessing the robustness, reliability, and user-friendliness of integrations all the more necessary, especially since the agricultural workplace has its own relevant set of data privacy concerns and industry standards.

## 2.6 Research gap and motivation

The existing literature has investigated the possibilities of blockchain and self-sovereign identity (SSI) integration in many different areas to some extent. Still, there is a lack of literature on their combined application in agricultural supply chains. Most of the current studies are either focused solely on blockchain or solely on SSI, and they do not examine the two technologies together. Current SSI frameworks largely originate from consumer identity management contexts and may not fully address the unique multi-stakeholder complexity of agricultural supply chains. That's a real gap because the agricultural supply chain has some unique needs: It needs to ensure data privacy, allow for identity verification, provide complete traceability to the end consumer, and decentralized governance. A theoretical framework that specifically addresses the power asymmetries and varied technical capabilities of agricultural stakeholders (from smallholder farmers to multinational distributors) is needed to guide future SSI implementations in this sector.

Conventional digital supply chain systems are dependent on centralized databases and identity management. Centralized identity management is inefficient and open to manipulation, and the limited transparency of centralized identity management databases can be detrimental to a supply chain's performance. While blockchain technology can substantially enhance supply chain transparency and traceability, it has some limitations. It does little to improve data privacy, identity ownership, or user control. Indeed, blockchain technology cannot handle these three critical functions because it is, at its core, a decentralized ledger that necessitates public visibility for it to function. [Table 1](#) summarise the research gaps identified for this study.

The complexity and interdependence of agricultural supply chains necessitate a holistic comprehension of blockchain and SSI integration. The very diversified nature of the various stakeholders that the supply chain comprises makes it challenging to develop coherent solutions. Thus, this study aims to make a comprehensive review of possible integration solutions. It will further help in answering the potential limitations and misconceptions. A second aim is to look for integration criteria that could allow us to evaluate the diverse technology solutions that are currently available. It will help the supply chain stakeholders make better decisions regarding which technologies to adopt. It will also guide future research toward the development of secure supply chain ecosystems that are operationally transparent and decentralized, with trust among different stakeholders.

## 3 Methodology

This systematic literature review (SLR) was carried out to investigate present blockchain and self-sovereign identity (SSI) solutions in agri-food supply chains. Its goal is to discover key trends, strengths, challenges, and areas of applicability for these two technologies within agricultural supply chains. This review focuses on answering three research questions that deal with the main components of blockchain and SSI technologies. This section explains the systematic search strategy used to identify articles relevant to the SLR.

## 3.1 Research questions

The main aim of this study is to conduct a systematic literature review of the use of blockchain and SSI solutions in agricultural supply chains. It examines current applications, identifies gaps in the literature, and aims to develop a comprehensive survey of the strengths and weaknesses of these technologies, particularly their applicability to the agricultural supply chain. The review's findings and analyses are intended to provide a better understanding of the limitations and advantages of these new technologies. The following set of research questions guides the review:

- **RQ1:** *What challenges arise in implementing blockchain within the agricultural supply chain sector?*
- **RQ2:** *What SSI solutions are available for agricultural supply chains?*
- **RQ3:** *What criteria are essential for comparing SSI solutions in agricultural supply chains?*
- **RQ4:** *What are the evaluation outcomes when comparing SSI solutions for agricultural supply chains?*
- **RQ5:** *What specific challenges and strengths characterize SSI solutions in agricultural supply chains?*

To address these guiding questions, we followed the systematic review guidelines outlined by ([Wohlin et al., 2020](#)) and standard procedures for literature selection to ensure a rigorous and comprehensive review process.

## 3.2 Search strategy

The search process involved the following steps:

- **Database Selection:** IEEE Xplore, Scopus, Web of Science, and ScienceDirect were chosen for their extensive collections of high-impact articles and advanced search functionalities.
- **Search Execution:** Each database was searched independently using the specified search strings.
- **Duplicate Removal:** Articles retrieved across multiple databases were filtered for duplicates using Mendeley software.
- **Preliminary Screening:** Titles, abstracts, and keywords were screened for relevance based on predefined inclusion and exclusion criteria.
- **Full-Text Review:** Articles passing the initial screening were reviewed in full to ensure alignment with the study objectives.

To thoroughly meet the research aims, a methodical search was conducted in four major scientific databases—IEEE Xplore, Scopus, Web of Science, and ScienceDirect. The intention was to find relevant literature concerning not only blockchain and decentralized identities but also their application in the agricultural supply chain. Searches were performed using a variety of keyword combinations to ensure exhaustive coverage of the relevant studies. The search strings themselves were created with extreme care. They reflect not only the core concepts of the study—blockchain, decentralized identity, and agriculture—but also their varied integration. Thus, the identified studies address not only the central concept but also several dimensions of it:

TABLE 1 Literature gap analysis.

| Ref                                                              | Identified gap                                                     | Industry/research needs                                                                            | Proposed research direction                                                              |
|------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Tran et al. (2021)                                               | Privacy concerns with blockchain-based systems                     | Mechanisms to ensure data privacy while maintaining transparency                                   | Design privacy-preserving blockchain systems using ZKPs or differential privacy          |
| Cocco et al. (2021)                                              | High costs and infrastructure demand for blockchain                | Cost-effective blockchain solutions optimized for resource-constrained environments                | Explore lightweight blockchain models with low operational costs                         |
| (Cocco et al., 2021; Manoj et al., 2023; Satybaldy et al., 2024) | Interoperability challenges with SSI frameworks.                   | Standards for interoperability between SSI frameworks and agricultural platforms                   | Develop standardized APIs and protocols for SSI frameworks in agriculture.               |
| Satybaldy et al. (2024),<br>Tharatipyakul et al. (2022)          | Low adoption among smallholder farmers due to complexity           | Accessible and user-friendly SSI solutions for rural stakeholders.                                 | Create intuitive SSI platforms with mobile-friendly interfaces and multilingual support. |
| Hasan et al. (2023), Duarte et al. (2024)                        | Minimal focus on decentralized governance                          | Decentralized governance mechanisms to ensure fairness and transparency                            | Propose decentralized governance models leveraging blockchain and SSI principles         |
| Duarte et al. (2024), Wang et al. (2023)                         | Lack of empirical studies validating blockchain-SSI in agriculture | Real-world data to evaluate the performance and feasibility of blockchain-SSI frameworks.          | Conduct pilot studies in agricultural supply chains and analyze outcomes.                |
| Tsoukas et al. (2022), Hasan et al. (2023)                       | Lack of integrated solutions combining blockchain and SSI          | Unified frameworks for secure identity verification and traceability in agricultural supply chains | Develop blockchain-SSI integration models tailored to agricultural ecosystems.           |

- Decentralized identity management (SSI).
- Blockchain technology applications.
- Agricultural supply chains.
- Their combined potential to enhance transparency, security, and traceability.

The final Boolean string used is:

- “Self-Sovereign Identity” AND “Agriculture”
- “Blockchain” AND “Self-Sovereign Identity” AND “Agriculture”
- “Blockchain” AND “Agriculture” AND “Supply Chain”
- “Self-Sovereign Identity” AND “Agriculture” AND “Supply Chain”
- “Blockchain” AND “Self-Sovereign Identity” AND “Agriculture” AND “Supply Chain”

Each search term was chosen for its relevance to the integration of SSI and blockchain technologies in agricultural contexts, particularly supply chains. The combinations used Boolean operators (AND, OR) to capture a broad spectrum of studies while maintaining specificity. These combinations were specifically designed to capture a broad spectrum of studies addressing the integration of SSI and blockchain technologies within agricultural contexts, especially in supply chain applications. The search focused on articles published between 2018 and 2024, emphasizing peer-reviewed papers, conference proceedings, and relevant grey literature to ensure a comprehensive examination of the subject.

3.3 Data sources

For this systematic review, literature was collected from several scholarly databases, including Scopus, Web of Science, ScienceDirect, and IEEE Xplore, chosen for their extensive collections of peer-reviewed articles and advanced search

functionalities. These databases support complex search expressions using keywords, author names, and abstracts, allowing for a thorough and targeted search. These databases also provide multidisciplinary coverage, which is essential because SSI research spans computer science, identity management, agriculture, and distributed systems. Each database was searched independently using specified search terms, and the results were combined after duplicates were removed with Mendeley software. In total, 1,229 articles were retrieved, comprising 238 from IEEE Xplore, 312 from Scopus, 293 from Web of Science, and 386 from ScienceDirect. Table 2 summarizes the number of articles obtained from each database, noting that some articles appeared in multiple sources. The selected time frame of 2018–2024 was defined based on the following considerations:

- SSI as a formalized concept gained momentum only after 2018, following major releases of decentralized identity standards.
- Agricultural blockchain implementations began scaling around the same period, aligning with industry adoption trends.
- Earlier studies (before 2018) primarily focused on generic blockchain use cases without formalized SSI architectures.

3.4 Inclusion/exclusion criteria and quality assessment

To refine the initial search results, specific inclusion and exclusion criteria were applied to ensure that only relevant articles aligned with the scope of this study were retained. These criteria were systematically applied to the titles, abstracts, and keywords of each identified article to determine relevance. In cases where the title and abstract did not provide sufficient information, the full text was reviewed to confirm adherence to the inclusion and exclusion criteria. To ensure the robustness and reliability of the studies included in this systematic review, a set of

TABLE 2 Search string and results for scholarly databases.

| Source                                                                           | IEEE Xplore | Scopus | Web of science | Science direct |
|----------------------------------------------------------------------------------|-------------|--------|----------------|----------------|
| "Self-Sovereign Identity" AND "Agriculture"                                      | 44          | 38     | 62             | 73             |
| "Blockchain" AND "Self-Sovereign Identity" AND "Agriculture"                     | 37          | 74     | 59             | 54             |
| "Blockchain" AND "Agriculture" AND "Supply chain"                                | 79          | 83     | 93             | 159            |
| "SSI" AND "Agriculture" AND "Supply chain"                                       | 57          | 51     | 48             | 63             |
| "Blockchain" AND "Self-Sovereign Identity" AND "Agriculture" AND "Supply chain." | 21          | 66     | 31             | 37             |
| Total with duplicates                                                            | 238         | 312    | 293            | 386            |

quality assessment criteria was developed and applied during the full-text review stage. [Table 3](#) summarizes these criteria.

- **Relevance to Research Objectives:** The study must directly address at least one research question related to blockchain or SSI in agricultural supply chains.
- **Methodological Transparency:** The study should provide a clear description of its methodology, including data collection, analysis, and limitations.
- **Data Validity and Credibility:** The study must use credible data sources (e.g., peer-reviewed journals and empirical case studies) and provide sufficient detail on how data were obtained and analyzed.
- **Analytical Depth:** The study must demonstrate a thorough analysis of its subject matter, discussing both strengths and limitations in depth.

Each paper was evaluated against a Gestalt quality criterion of rigor, clarity of findings, and relevance to our research questions by two researchers independently, with discrepancies resolved through discussion. Those papers that were deemed to fall short were excluded from the final analysis. Grey literature (e.g., theses, preprints, opinion papers) was screened initially but removed at the eligibility stage because it did not meet peer-review standards. This ensured methodological rigor and consistency. Given that blockchain-based SSI implementation is a relatively new research area, incorporating grey literature was also considered to broaden the scope and capture emerging perspectives from industry and academia.

### 3.5 Screening and selection process

A total of 1,229 articles were collected from scholarly databases to ensure comprehensive coverage of the subject matter. To maintain a focus on academically rigorous sources, grey literature was intentionally added. This initial collection included duplicate entries, demanding a systematic approach to ensure uniqueness within the dataset. [Figure 2](#) illustrates this workflow, detailing the key phases undertaken during the refinement process. Following this step, 331 duplicate records were identified and removed, resulting in a dataset of 898 unique articles, which helped eliminate redundancy that could cause bias in the study.

TABLE 3 Inclusion/exclusion criteria.

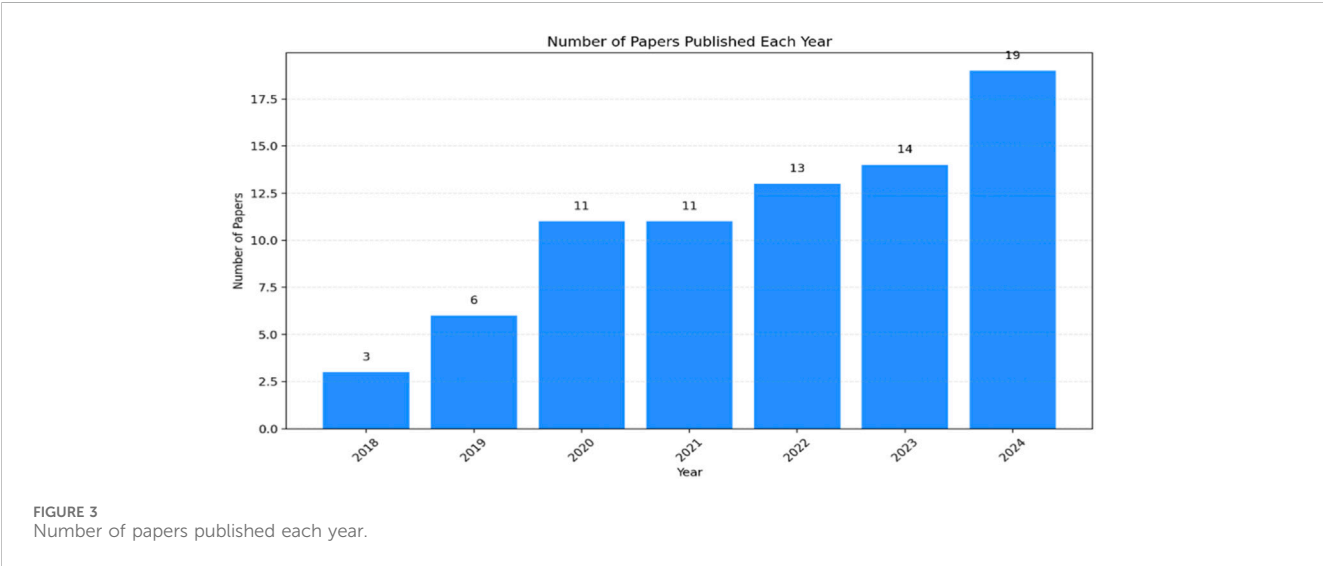
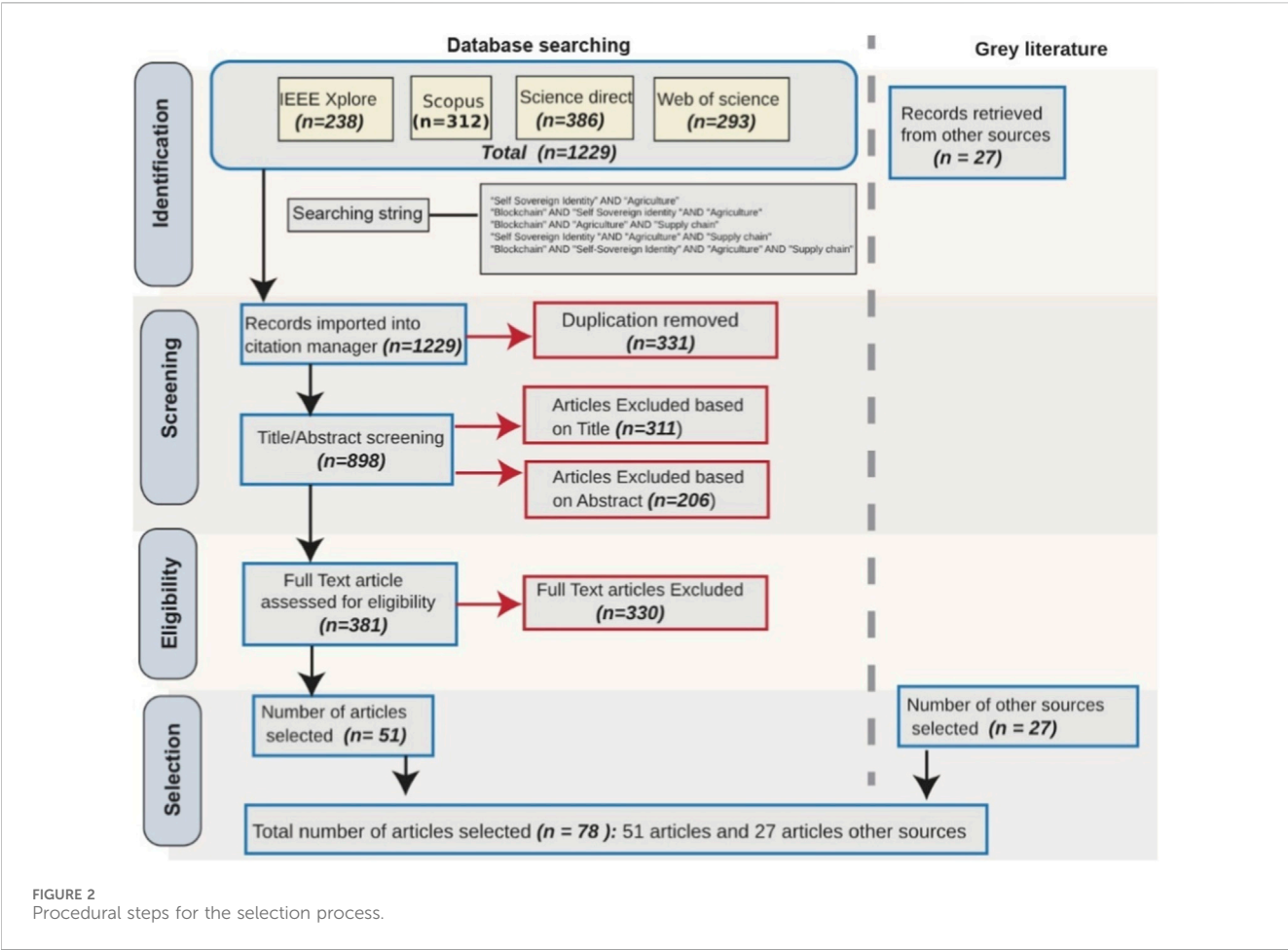
| Inclusion criteria                                                                   | Exclusion criteria                                                             |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| i. Articles published between 2018 and 2024                                          | i. Duplicate studies                                                           |
| ii. Studies focused on blockchain and SSI applications in agricultural supply chains | ii. non-English language publications                                          |
| iii. Original research articles (not reviews or surveys)                             | iii. Papers with content unrelated to blockchain-based self-sovereign identity |
| iv. English language publications                                                    | iv. non-peer-reviewed articles (e.g., editorials, opinion pieces)              |
|                                                                                      | v. Articles not relevant to SSI, blockchain, or agriculture                    |
|                                                                                      | vi. Quality assessed holistically based on rigor and clarity of findings.      |

The annual distribution of academic papers published between 2018 and 2024 is shown in [Figure 3](#). The data started with 03 papers in 2018 and moved through a modest but consistent number of titles up to 19 papers in 2024. The percentage distribution of academic papers published each year from 2018 to 2024 is shown in [Figure 4](#). From these data, it is clear that 2018 was the year with the lowest percentage of publications, with just 3.9% of the total number of publications coming out that year. After that, there was a steady climb in the number of publications. There were a couple of years where the number stayed the same (2020 and 2021) at 14.3%. The major year by far in this study was 2024, with 24.7% of the total number of publications that year. So, this shows us not just steady growth over the period of this study but also a really significant research interest in this topic in 2024. [Table 4](#) presents the summary of regional case studies in the domain of study.

[Figure 5](#) illustrates the cumulative growth of publications from 2018 to 2024. Here's a breakdown of the data:

Key observations from the graph:

1. **Steady Growth:** The graph shows a consistent upward trend, indicating a steady increase in the cumulative number of publications over the years.
2. **Year-by-Year Growth:**
  - 2018: Started with just 03 papers.
  - 2019: Increased to six papers (3 new papers).



- 2020: Reached 11 papers (5 new papers).
  - 2021: Stabilized at 11 papers (no additional growth).
  - 2022: Increased to 13 papers (2 new papers).
  - 2023: Grew to 14 papers (1 new paper).
  - 2024: Jumped significantly to 19 papers (5 new papers).
3. Highest Increases: The most notable growth occurred between 2023 and 2024, with 05 additional papers published, indicating a recent surge in research activity.

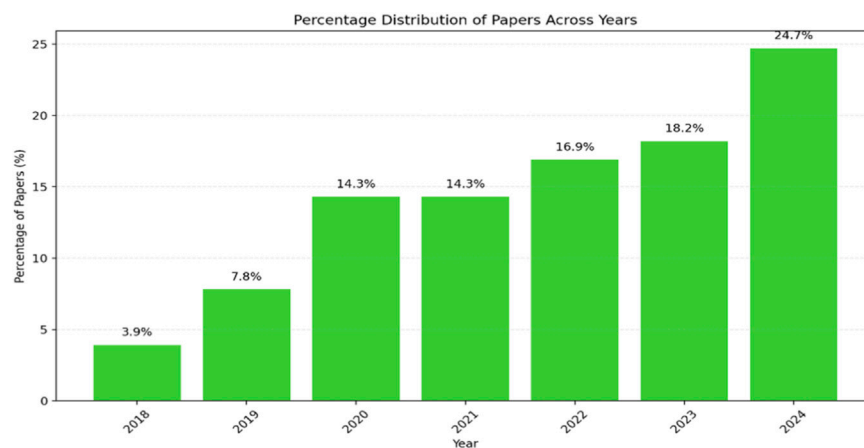


FIGURE 4  
Percentage distribution of papers across years.

TABLE 4 Summary of regional case studies dealing with blockchain application in food supply chain.

| Ref                                  | Country                  | Initiative name                   | Sector                      | Blockchain application                                     | Features                                                                                             |
|--------------------------------------|--------------------------|-----------------------------------|-----------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Mohammed et al. (2024)               | Ghana                    | Cocoa Traceability Initiative     | Cocoa                       | Supply chain transparency                                  | Enhances ethical sourcing and transparency in cocoa supply chains                                    |
| Kamath (2018)                        | United States of America | IBM Food Trust                    | General Food                | Food safety and traceability                               | Improves food safety and reduces waste by providing end-to-end visibility of the food supply chain   |
| Gulen et al. (2024)                  | Turkey                   | Turkish Agricultural Blockchain   | General Food                | Traceability and quality assurance                         | Aims to improve transparency and traceability in food supply chains                                  |
| Peng et al. (2022)                   | China                    | Smart Agriculture Blockchain      | General Food                | Agricultural finance and traceability                      | Integrates financial services with supply chain transparency                                         |
| Mishra and Maheshwari (2021)         | India                    | Blockchain in Public Distribution | Food Security               | Prevents supply leakage                                    | Enhances transparency and reduces corruption                                                         |
| Premchandran and Batchu Veera (2023) | India                    | AgriChain                         | General Agriculture         | Supply chain management                                    | Improves efficiency and reduces waste                                                                |
| Rambhia et al. (2022)                | India                    | Agri10x                           | Agriculture                 | Marketplace Platform                                       | Connects farmers directly with buyers, ensures transparency                                          |
| Xu et al. (2019)                     | Australia                | AgriDigital                       | Grains                      | Digital grain trading                                      | Facilitates secure transactions and ownership                                                        |
| Kamilaris et al. (2019)              | Netherlands              | Provenance                        | Various Agricultural        | Product origin tracking                                    | Empowers consumers with information about product origins                                            |
| Kumar and Perepu (2023)              | Kenya                    | Twiga Foods                       | Fresh Produce               | Supply Chain Optimization                                  | Connects farmers directly with retailers                                                             |
| Ndlovu and Masuku (2021)             | South Africa             | Grain SA                          | Grains                      | Market access for farmers                                  | Enhances market transparency and farmer income                                                       |
| Li et al. (2023b)                    | Brazil                   | BlockAgro                         | General Agriculture         | Traceability in agricultural products                      | Combats fraud and ensures quality                                                                    |
| Thiruchelvam et al. (2018)           | Burundi                  | Coffee Blockchain                 | Coffee Supply Chain         | Coffee traceability and fair trade                         | Ensures transparency, traceability, and fairness for coffee producers and exporters                  |
| Nyuk Mee et al. (2021)               | Malaysia                 | Agrolink                          | Agriculture and Food Supply | Information sharing platform for agricultural stakeholders | Links farmers, producers, service providers, and institutions for seamless supply chain connectivity |

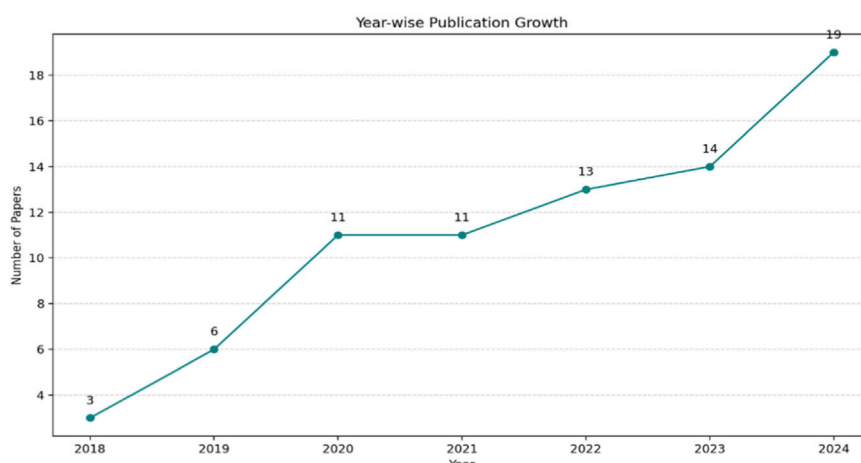


FIGURE 5  
Research papers year-wise cumulative growth.

4. Overall Output: By the end of 2024, a total of 19 papers were published, which is more than six times the number published in 2018, reflecting substantial cumulative progress.

This cumulative growth graph effectively shows the overall trend of the research output over the 7 years. It reveals not just the year-by-year variations but also the total amount of work that has been done in this field. The remaining 898 articles were then subjected to a preliminary review using their titles, abstracts, and keywords. This initial screening was intended to keep directly relevant articles while excluding those not aligned with the research's main thrust. The criteria for this screening were straightforward: If the article was about something too far from the field, or if the study's keywords did not match the article's title, the article was removed from the pool. A total of 311 articles were removed because their titles did not indicate enough relevance, and a further 206 were rejected after their abstracts were reviewed. The remaining articles were subjected to a more thorough evaluation that involved closely examining their abstracts, discussions, and conclusion sections. The criteria for inclusion emphasized selecting articles that contained either empirical data, theoretical insights, or any relevant case study aligned with the study's objectives. After this process, the number of "selected" articles was reduced to 381. The total of 381 articles was then filtered down through a full reading of their content to the 78 articles that form the core of the study.

## 4 Results

### 4.1 Research question 1: What challenges arise in implementing blockchain within the agricultural supply chain sector?

1. Agricultural supply chain sector case studies.

The integration of advanced technologies—such as blockchain and IoT—has brought the agricultural supply chain a number of

favorable changes. These changes have improved transparency, efficiency, and traceability in the supply chain's operations. Case studies and research findings provide a good view of the overall situation, allowing us to see how technological impact benefits the supply chain and what challenges we need to overcome for better implementation. The decentralized, tamper-proof nature of blockchain could solve long-standing issues of trust and transparency in supply chains (Hasan et al., 2023). Mention that when farms, factories, and other supply chain stakeholders are linked *via* the Internet of Things, visible traces are left all along the path that food travels. With the right technology, that path can be followed virtually in real-time. Following that virtual path could and should eliminate danger and fraud from the business of tricking food consumers and ensure that all food stakeholders are appropriately informed. This seamless, virtual following of the food path could be what guarantees food safety and stakeholder trust (Zhang et al., 2021). say that it is in following this path that one can see the trust and transparency that blockchain provides to food supply chains.

#### 4.1.1 Regional case studies

This section provide the review of blockchain applications in various countries/regions for agri supplychain.

- Ghana: Initiatives using blockchain technology are progressing toward responsible sourcing in Ghana's cocoa and agricultural sectors (Mohammed et al., 2024). Applying the Diffusion of Innovation Theory, determined that the adoption of blockchain fosters ethical sourcing practices and raises the level of transparency within the cocoa industry—so vital to Ghana's economy—a necessary condition for significantly improving the level of accountability within the industry.
- United States: Walmart's alliance with IBM on the Food Trust blockchain platform serves as an excellent case study of the potential for food safety and traceability. This initiative allows stakeholders to ensure that the food we eat is safe and traceable—right from the farm to our dining room tables.

And with Walmart's scale, it is difficult to imagine a more compelling test case (Traceability is not as relevant, of course, for the food that is inside the typical dining room fridge or freezer (Kamath, 2018)).

- Turkey: Turkey is considering blockchain to boost transparency in the food supply chain from production to delivery (GULEN et al., 2024).say that working together, government and private entities can make the IBM Food Trust a feasible solution for managing food supply chains.
- China: Smart Agriculture in China applies blockchain to the rice supply chain. Enhancing traceability means fighting against counterfeit products. In this way, it maintains the quality that is expected from food grains (Peng et al., 2022).
- India: Several different initiatives in India showcase blockchain's versatility:
  - i.The Public Distribution System (PDS) seeks to slash leakage and corruption in the food distribution system by implementing a consortium blockchain that aims to achieve transparency and accountability in the system (Mishra and Maheshwari, 2021).
  - ii.Aagri10x establishes a direct connection between farmers and buyers that bypasses intermediaries, ensuring that farmers receive fair, market-driven prices for their products. At the same time, it maintains the integrity of the food supply system (Premchandran and Batchu Veera, 2023).
  - iii.AagriChain is primarily focused on providing supply chain efficiencies that minimize wastage and ensure that food integrity is maintained at every stage of the supply chain (Rambhia et al., 2022).
- Australia: A public blockchain platform, AgriDigital, enables secure digital trading of grain and enhances transaction security and ownership verification in the grain supply chain. This company is working on the grain problem, which is primarily about (supply chain) efficiency and transaction security (Xu et al., 2019).
- Netherlands: Provenance is a large-scale initiative in the Netherlands. It is aimed at establishing transparency in the agricultural sector and making it possible for consumers to trace the origins of their food. It has utilized public blockchain technology to create a decentralized and accessible system of records (Kamilaris et al., 2019).
- Kenya and South Africa: In Kenya, Twiga Foods uses a private blockchain to link farmers with retail trade to connect the produce of farmers to the customers who consume it. Twiga (a Swahili word for "giraffe") was launched to ensure the market fairly connects to those farmers (Kumar and Perepu, 2023). In a similar vein, Grain SA uses consortium blockchain technology to connect farmers to retail and wholesale markets in a more transparent and trustworthy way. Farmers who grow grain should be able to find better markets through this technology (Ndlovu and Masuku, 2021).
- Vietnam: (Tsai et al., 2023): highlight the potential of blockchain technology in Vietnam's agricultural sector. They identify barriers to adoption, such as a lack of regulation, scalability, and resource requirements. Their study proposes a hierarchical approach to address these challenges and facilitate the integration of blockchain into the agricultural sector.

- Burundi: The coffee supply chain, where conventional methods struggle with traceability and visibility, could benefit from blockchain (Thiruchelvam et al., 2018). Apply the Technology Acceptance Model (TAM) to advocate for an equitable, automated trade system that would enhance efficiency and trust in the coffee sector of this nation.

While these case studies highlight blockchain's potential, most of them focus on traceability and product movement rather than identity assurance. Also, only a few have integrated SSI principles. For example, Walmart's Food Trust implementation focuses primarily on product traceability rather than stakeholder identity management. It highlights the need for solutions that combine the traceability benefits of blockchain with the privacy and control advantages of SSI frameworks. This gap reinforces the need to integrate SSI to authenticate stakeholders as well as products, as the regional case studies collectively demonstrate that current implementations overwhelmingly prioritize product tracking while offering little to no support for decentralized identity verification—the core challenge SSI is intended to address.

#### 4.1.2 Summary of implementation challenges

Although blockchain technology offers many benefits to agricultural supply chains, such as increased transparency, enhanced trackability, and augmented security, it has some limitations:

- Integration and Standardization: Scalability is hampered by the complex integration requirements and the absence of standardized frameworks (Kamilaris et al., 2019; Nagariya et al., 2022). Observe that integrating blockchain with pre-existing agricultural systems necessitates profound infrastructural transformations.
- Cost and Technological Infrastructure: Adoption is primarily hindered by high costs and the need for a robust infrastructure, both of which are huge burdens on the limited digital resources seen in many parts of the world as (Cuellar and Johnson, 2022) highlight these as serious obstacles and recommend government support, investment, and education of stakeholders involved in these areas to overcome them.
- Privacy and Data Management: Ensuring the privacy of large volumes of data while managing them remains a challenge (Anjum et al., 2024). have proposed a storage model for on-chain and off-chain data that could help with achieving this balance. It is essential for sectors with large data loads, like the fruit and vegetable sector.
- Regulatory and Cultural Barriers: The adoption of the blockchain in different areas is being held back by inconsistencies in regulatory frameworks, as well as cultural and technological disparities. Research conducted in East Africa (Omanwa, 2023) and Vietnam (Vu and Trinh, 2021) indicates that it is vital to establish clear local regulations and to significantly enhance the digital literacy of stakeholders in order to facilitate the disruptive potential of the blockchain.
- Scalability and Energy Consumption: Huge energy needs and limited scalability can hinder blockchain's sustainability in large-scale applications (Ellahi et al., 2023). recommend the exploration of hybrid models and integration of the

technologies of Industry 4.0 and Web 3.0 to make the blockchain more efficient.

## 4.2 Research question 2. what SSI solutions are available for agricultural supply chains?

SSI offers a user-centric approach to digital identity management, empowering individuals to control their personal data without reliance on centralized authorities. SSI's decentralized framework is highly relevant to agricultural supply chains, where transparency, traceability, and trust are crucial. This section reviews available SSI solutions and explores how they can address identity management needs within agricultural contexts.

### 4.2.1 Key SSI solutions and their applications

- **Sovrin:** Designed with a specific purpose—to create a next-generation public, permissioned blockchain for decentralized digital identities. Users can and do control their data, and they do it in a direct manner that accommodates the kind of transactions necessary to move an agriculture-based economy towards one that maintains user privacy (Naik and Jenkins, 2020; Sousa et al., 2022). For agricultural supply chains, Sovrin's support for verifiable credentials and selective disclosure can help authenticate farmers, certifiers, and logistics providers while preserving privacy.
- **uPort:** uPort runs on the public Ethereum blockchain, providing a decentralized, open-source SSI system that lets users control and store their credentials in an app on their mobile devices. This credential processing system enables selective information sharing, which is essential in agriculture. For example, if a stakeholder wants to see your certifications (organic, fair trade, etc.), you also want to maintain a certain level of privacy (Kondova and Erbguth, 2020). However, transaction costs and Ethereum's scalability constraints may affect adoption in large or resource-constrained agricultural networks.
- **SelfKey:** SelfKey highlights user control by enabling individuals to disclose information to service providers on a selective basis. It aims to make service provision and governance more secure and user-friendly while removing the dependencies on many current centralized authorities. Similarly, in agricultural supply chains, SelfKey could help governance "providers" (like regulators, certifiers, etc.) to find the crucial information they need from individuals without a dependency on a centralized authority (Giannopoulou and Wang, 2021).
- **ShoCard:** ShoCard utilizes blockchain technology to decentralize identity management and allows users to create and manage their digital identities securely. Its model could apply to minting and managing digital certificates in agriculture; using ShoCard as an example, we might consider how the model could be used to enable direct verification of agricultural product authenticity when users are presented with a digital certificate (Haddouti and Ech-Cherif El Kettani, 2019).
- **Civic:** Civic leverages blockchain technology to grant users command over their virtual identities. They provide only the most necessary personal data and only when essential (Acharya et al., 2023). This scheme fits well with the way the agricultural supply chain should work by disclosing only the most necessary information without compromising the privacy of the entities involved.
- **Blockstack:** A decentralized identity system is offered by Blockstack. Blockstack uses blockchain-based Decentralized Identifiers (DIDs) to enable identity verification across multiple applications. This platform is known for providing a unified identity across numerous applications; thus, it could simplify interactions across the agricultural supply chain by furnishing consistent identity verification for different stakeholders (Ali et al., 2017; Rathee and Singh, 2022). At the same time, its technical requirements may pose challenges for smallholder farmers or low-resource cooperatives.
- **Evernym:** Evernym, constructed on the Sovrin network, offers decentralized identity tools that enable secure identity verification (Rathee and Singh, 2022). When it comes to the agriculture domain, Evernym could help facilitate real-time, secure identity verification among supply chain actors, which might enable them to trust one another more and rely less on intermediaries.
- **EverID:** EverID concentrates on secure identity verification and asset transfer, which could enhance the efficiency of agricultural supply chains by requiring fewer intermediaries. This approach could be especially beneficial for small-scale farmers who want to access supply chain networks (Dib and Toumi, 2020; Hünseler and Pöll, 2023). Nonetheless, deploying such systems at scale requires adequate digital infrastructure and user training, which may not always be available.

### 4.2.2 Role of SSI in enhancing agricultural supply chains

The technologies underpinning Self-Sovereign Identity (SSI) can enhance traceability, transparency, and efficiency within the agricultural supply chain. When stakeholders use SSI, they rely on decentralized identifiers (DIDs) and verifiable credentials to authenticate themselves and to access the certifications that allow them to participate in the supply chain. These mechanisms do not depend on centralized authorities and thus are much closer to the ideal of "decentralized, permissionless, and trustless blockchain systems." Accordingly, using SSI to create a blockchain-based supply chain can offer stakeholders more security and privacy and give them more control (Lyu et al., 2022). In practical agricultural contexts, this means that farmers, cooperatives, logistics providers, and certifiers can present verifiable credentials (e.g., organic labels, Good Agricultural Practices certificates) in a privacy-preserving way, while downstream buyers and regulators can verify these claims without accessing unnecessary personal data.

### 4.2.3 Challenges and principles for SSI adoption

SSI faces several challenges:

- **Integration and Interoperability:** Different SSI implementations often lack seamless integration across systems. Efforts towards standardization and

interoperability are crucial for SSI's success in global supply chains.

- **Complexity and Accessibility:** SSI solutions can be technologically complex, presenting challenges for stakeholders with limited digital literacy, such as smallholder farmers.
- **Regulatory Compliance:** Compliance with international data protection standards, such as GDPR, remains a significant hurdle for many SSI solutions (Grüner et al., 2023).

To address these challenges, SSI systems should adhere to foundational principles of identity management, including:

- **User Control and Consent:** Ensuring users control their identity data.
- **Privacy and Minimum Disclosure:** Allow only limited data sharing, which is essential for each transaction.
- **Decentralization:** Avoiding dependence on central authorities, thus reducing data security risks.
- **Transparency and Interoperability:** Creating an open ecosystem where SSI solutions can function harmoniously across diverse supply chain systems.

Sovrin is offering an approach to digital identity management within agricultural supply chains that is truly transformative and tackles the three “Ts” of transparency, traceability, and stakeholder empowerment. Sovrin, along with its uPort and Civic, is a strong proponent of decentralized digital identity, which is suitable for the essential requirements of secure, verifiable claims in the agriculture domain. It empowers all parties throughout that supply chain to know who or what they are dealing with at each step.

### 4.3 Research question 3: What criteria are essential for comparing SSI solutions in agricultural supply chains?

#### 4.3.1 SSI principles

User-centric electronic identity systems are becoming increasingly popular worldwide. These systems aim to give individuals more control over their digital identities. Different concepts of how identities could be managed in digital environments have emerged over the last 2 decades. Probably the most popular concept today is self-sovereign identity (SSI) (Schardong and Custódio, 2022; Satybaldy et al., 2024). Along with the clear advantages and relevance, there are some challenges and obstacles to SSI implementation. The guiding principles of SSI implementation are:

- **User Control and Consent:** It allows all kinds of stakeholders, including farmers and distributors, to manage their identity data, especially the personal and sensitive information about their farming practices, certifications, and what they produce. It ensures that identity data management promotes trust because data-sharing happens only when identity owners give their consent.
- **Privacy and Protection:** This ensures the data's confidentiality. It allows users to protect information tied to their proprietary

farming practices and methods. It protects user data all the way through the supply chain, reducing unauthorized access and competitors' prying eyes.

- **No Trust in Central Authority:** Advocates for a decentralized model—no single entity is in charge. This lack of top authority means that farmers, distributors, and retailers can trust one another. In fact, they have to trust one another because there is not an overarching authority to step in if something goes wrong.
- **Transparency:** Providing supply chain data to all participants permits complete traceability. Openness also allows auditing, which is essential for tracking how products move through the supply chain.
- **Interoperability:** Ensuring compatibility between diverse platforms, databases, and standards is essential for a vast network like the agricultural supply chain, where many stakeholders interact across different sectors. Interoperability allows these stakeholders to work together and share information smoothly.
- **Scalability:** An SSI system's scalability assures that it can handle the high number of participants and transactions that occur in agricultural supply chains.
- **Usability:** SSI prioritizes the ease and simplicity of use for farmers and small-scale producers who are not technical. It ensures that the solutions advanced by SSI are clear, understandable to support widespread adoption among diverse agricultural stakeholders.
- **Portability and Persistence:** Ensuring that users can maintain their digital identity across various platforms and service providers is a basic tenet of good design. Portability is crucial for agricultural stakeholders, who may work with a variety of services.

#### 4.3.2 Evaluation principles for comparing SSI solutions

Researchers have created several criteria to evaluate SSI solutions based on foundational identity principles. Table 5 presents the most significant of these principles, which are used in comparative studies of SSI solutions. Allen's ten SSI principles are often used to define criteria for such evaluations (Liu et al., 2020). However, the authors also evaluate the respective SSI systems based on their functionalities. These studies highlight control, minimal disclosure, and interoperability as the most important criteria for the evaluation of SSI systems. In agricultural contexts, these criteria are particularly relevant because identity verification must be secure, lightweight, and compatible across many actors, including smallholder farmers, cooperatives, regulators, and exporters.

#### 4.3.2 Key SSI principles for the agricultural supply chain: focusing on four key criteria

The agricultural supply chain encounters a range of challenges that need specific solutions for identity management. These challenges center on three key aspects: assuring traceability, transparency, and security in an environment with many stakeholders of varying capability and trust levels. Consequently, we identified four core principles of Self-Sovereign Identity that directly address these needs and potentially improve the overall efficiency of the supply chain, build trust among its members, and

TABLE 5 Key SSI principles addressing agricultural supply chain challenges.

| SSI principle                    | Reason for selection                                                                                                                                                                                                                                                                                                                                                | Addresses challenges of      |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1. Transparency                  | <b>Enhances Traceability:</b> It allows participants to verify identity claims and transactions, which is essential for tracking products from farm to table.<br><b>Builds Trust:</b> Provides an auditable ledger (e.g., blockchain), fostering trust among stakeholders.<br><b>Directly Tackles Traceability:</b> Makes the supply chain visible and accountable. | Traceability<br>Transparency |
| 2. Privacy and Protection        | <b>Secures Sensitive Data:</b> Protects personal and business information from unauthorized access, maintaining confidentiality.<br><b>Maintains Competitive Advantage:</b> Safeguards proprietary methods and data.<br><b>Enhances Security:</b> Prevents data breaches and manipulation.                                                                          | Security                     |
| 3. No Trust in Central Authority | <b>Eliminates Single Points of Failure:</b> Reduces risks associated with central authorities being compromised.<br><b>Promotes Equal Participation:</b> Prevents monopolistic control over data.<br><b>Strengthens Security:</b> This makes system manipulation by malicious actors more difficult.                                                                | Security                     |
| 4. User Control and Consent      | <b>Empower Stakeholders:</b> Users control their identity data, deciding what to share and with whom.<br><b>Facilitates Compliance:</b> Aligns with data protection and privacy regulations.<br><b>Supports Transparency and Security:</b> Encourages active, responsible participation, maintaining data integrity.                                                | Transparency<br>Security     |

ensure resilience. Although SSI literature commonly outlines six foundational principles, this review narrows the evaluation to four because these criteria—transparency, privacy and protection, no trust in a central authority, and user control and consent—most directly map onto the identity-related challenges observed in agricultural supply chains. The remaining two principles, while theoretically relevant, were found to have limited operational significance in this specific context. These four core principles, summarized in Table 5, therefore provide a focused and context-appropriate lens for comparing SSI solutions in agricultural environments.

#### 4.3.2.1 Transparency

- **Rationale:** Transparency is necessary to achieve traceability. This is particularly true for agricultural supply chains because they require working systems that allow for the tracking of products from their origins to the consumers who buy them. Too often, the agricultural sector has to deal with supply chains that are contaminated, fraudulent, or poorly labeled. These supply chains lack transparency, and processes are opaque. In contrast, the optimum supply chain works with a known network of stakeholders who are transparent and verifiable.
- **Agricultural Context:** With transparency, an SSI system can provide an immutable audit trail, which enhances trust between parties. Every transaction that takes place is available for verification. This is especially important in the food industry and regulatory compliance. It is essential for protecting consumers and ensuring that businesses meet the standards set by the various regulators that oversee our food supply.

#### 4.3.2.2 Privacy and protection

- **Rationale:** The protection and privacy of sensitive data, like farming methods, supplier details, and consumer information, are crucial.

- **Agricultural Context:** Ensuring privacy allows the parties involved to disclose information selectively, thereby enabling them to minimize the risks that come with exposing their data.

#### 4.3.2.3 No trust in central authority

- **Rationale:** The supply chains for agriculture are made up of many actors, often spread out and decentralized, who participate in the different stages of the chain from farm to customer.
- **Agricultural Context:** Promoting a trustless environment makes data verification completely decentralized, eliminating ties to any central authority that might fail or be abused.

#### 4.3.2.4 User control and consent

- **Rationale:** Fostering trust in situations where stakeholders are hesitant to share information without retaining control requires ensuring user control and consent. Data ownership is a sensitive issue in agricultural supply chains, particularly when the data pertains to farmers' individual farming practices or proprietary growing methods.
- **Agricultural Context:** This principle guarantees that farmers, suppliers, and other stakeholders have complete authority over their data. They decide when and with whom to share it. Control encourages participation. It assures users that their data will not be accessed or used without their consent.

### 4.4 Research question 4: What are the evaluation outcomes when comparing SSI solutions for agricultural supply chains?

Prominent SSI solutions have been compared based on the key principles that are essential for effective implementation in

agricultural supply chains. These key principles are based on the attributes necessary for identity management systems in the agricultural supply chain: They need to be secure; they need to enable verifiable interactions; and they need to allow for the decentralized management of identities—basically, they need to be trusted. This comparison is necessary because it will enable us to identify which SSI solutions can be effectively used in the context of the agricultural supply chain.

i. **Sovrin** (Windley, 2021)

- User Control and Consent: Sovrin empowers users to have full control over identity data. Farmers and other agricultural stakeholders can selectively disclose information using Zero-Knowledge Proofs (ZKPs) (Naik and Jenkins, 2021).
- Privacy and Protection: Advanced cryptographic methods like ZKPs protect data, making Sovrin highly secure for storing sensitive agricultural data (Shuaib et al., 2022b).
- No Trust in Central Authority: Sovrin operates on a fully decentralized framework, eliminating reliance on a central authority.
- Transparency: The platform's blockchain architecture ensures immutable records, supporting traceability from production to distribution.

ii. **uPort** (Naik and Jenkins, 2020)

- User Control and Consent: Built on Ethereum, uPort allows stakeholders in agricultural supply chains to retain ownership of data *via* a mobile application.
- Privacy and Protection: Privacy is maintained through user-controlled cryptography, though some technical complexity may limit usability.
- No Trust in Central Authority: Ethereum's decentralized structure aligns with uPort's aim to eliminate central authorities.
- Transparency: Blockchain-backed transparency supports credential verification, though but high gas fees and congestion may limit adoption in resource-constrained agricultural settings (Grüner et al., 2023).

iii. **SelfKey** (Foundation, 2017)

- User Control and Consent: SelfKey prioritizes user control, enabling individuals to manage data, which is especially beneficial for stakeholders handling sensitive supply chain certifications (Pradhan and Shah, 2025).
- Privacy and Protection: Data minimization and selective disclosure features protect user privacy.
- No Trust in Central Authority: SelfKey's decentralized nature reduces dependency on intermediaries.
- Transparency: Transparent, blockchain-based transactions enhance trust by allowing participants to verify data authenticity.

iv. **Civic**

- User Control and Consent: Civic gives users control over personal data, which is a benefit in agricultural supply chains for managing certification and production data (Pava-Díaz et al., 2024).
- Privacy and Protection: Civic uses Zero-Knowledge Proofs and selective disclosure for privacy, but the complexity may present challenges in managing agricultural data.

- No Trust in Central Authority: Operates independently of a central authority, ensuring trustless interactions.
- Transparency: Blockchain's immutable ledger enhances transparency, which is critical for traceability.

v. **Blockstack** (Ali et al., 2016)

- User Control and Consent: This option offers user-managed cryptographic keys, though usability concerns may limit accessibility in non-technical agricultural settings.
- Privacy and Protection: Provides privacy through selective disclosure, but complexity may impact practical application (Ahmed et al., 2022).
- No Trust in Central Authority: Decentralization aligns well with trustless interactions, reducing the risks of single points of failure.
- Transparency: Blockchain integration provides a clear, immutable audit trail, enhancing traceability.

vi. **ShoCard**

- User Control and Consent: While users control their data, ShoCard's technical complexity may make it less accessible to non-technical stakeholders in agriculture.
- Privacy and Protection: Relies on cryptographic security to maintain privacy but lacks some of the advanced privacy features of other solutions (Pava-Díaz et al., 2024).
- No Trust in Central Authority: Decentralized, removing reliance on a central authority and aligning with SSI principles.
- Transparency: Blockchain facilitates transparent interactions, though the lack of specialized support for agricultural transparency limits its use. But its general-purpose design offers limited direct support for agricultural traceability (Shadung et al., 2024).

vii. **LifeID**

- User Control and Consent: Offers a strong model for user control, allowing agricultural stakeholders to manage access to data selectively (Pradhan and Shah, 2025).
- Privacy and Protection: Selective disclosure through advanced cryptographic methods supports privacy, which is crucial for handling sensitive agricultural information.
- No Trust in Central Authority: Operates without a central authority, using decentralized ledgers for trustless data verification.
- Transparency: Blockchain provides an immutable record of transactions, allowing stakeholders to trace data across the supply chain.

viii. **Evernym**

- User Control and Consent: Provides complete user control, allowing stakeholders to manage and selectively disclose credentials.
- Privacy and Protection: Decentralized identifiers and verifiable credentials protect privacy, an asset for maintaining confidentiality in supply chain data.
- No Trust in Central Authority: Decentralized infrastructure supports trustless interactions across agricultural networks.
- Transparency: Blockchain's transparency aligns well with agricultural requirements for traceability, allowing end-to-end product tracking.

TABLE 6 Comparative study of the blockchain-based SSI solutions.

| SSI solutions | SSI principles (evaluation criteria) |                        |                               |              |
|---------------|--------------------------------------|------------------------|-------------------------------|--------------|
|               | User control and consent             | Privacy and protection | No trust in central authority | Transparency |
| Sovrin        | ✓                                    | ✓                      | ✓                             | ✓            |
| uPort         | ✓                                    | ✗                      | ✓                             | ✗            |
| SelfKey       | ✓                                    | ✓                      | ✓                             | ✗            |
| Civic         | ✗                                    | ✗                      | ✓                             | ✗            |
| Blockstack    | ✗                                    | ✗                      | ✓                             | ✗            |
| ShoCard       | ✓                                    | ✗                      | ✓                             | ✗            |
| LifeID        | ✓                                    | ✓                      | ✓                             | ✓            |
| Evernym       | ✓                                    | ✓                      | ✓                             | ✓            |
| EverID        | ✗                                    | ✗                      | ✓                             | ✗            |

✓ indicates substantial compliance with the principle; ✗ indicates significant limitations in addressing the principle. Assessment based on published literature and technical documentation.

#### ix. EverID

- **User Control and Consent:** This option empowers users to manage identity information directly, which is a plus for agricultural stakeholders who manage their data independently.
- **Privacy and Protection:** This option offers privacy controls, though these may be more complex and harder to manage for non-technical users.
- **No Trust in Central Authority:** The decentralized architecture avoids reliance on centralized authorities, reducing control risks.
- **Transparency:** Provides some transparency through blockchain, but complexity and lack of specialized agricultural features may limit its effectiveness.

Table 6 shows that no SSI solution achieves full compliance across all four principles. Sovrin, Evernym, and LifeID exhibit the strongest overall alignment, performing well in decentralization, privacy, and user control. In contrast, Civic, Blockstack, and EverID show weaker performance in privacy and user empowerment, indicating limitations for sensitive agricultural data environments. This comparison reveals that highly decentralized and privacy-focused solutions tend to score better for agricultural supply chain requirements, whereas systems lacking selective disclosure or strong user-centric controls are less suitable. An in-depth analysis of SSI solutions based on these four essential principles concludes that Sovrin is the most compatible and effective solution for agricultural supply chains. With its decentralized structure, state-of-the-art cryptographic safeguards, transparent blockchain record-keeping, and, most importantly, its user-controlled data-sharing capability, Sovrin is uniquely qualified and built to manage identity across agricultural networks. It provides:

- **Complete Decentralization:** Sovrin's decentralized structure promotes peer-to-peer trust, which is essential in a sector with varied and dispersed entities.

- **Strong Privacy:** Sovrin achieves a high level of privacy using zero-knowledge proofs and selective disclosure. It is essential as agricultural data can be highly sensitive.
- **User Empowerment:** Sovrin's user-centric design empowers not just one but all stakeholders because they not only share trusted, direct interactions with one another but also have full control over their data.
- **Immutability and Auditability:** Sovrin's blockchain foundation provides transparency. All the reliable features of the blockchain can be used to create a record of interactions, making the record auditable and verifiable.

## 4.5 Research question 5: What specific challenges and strengths characterize SSI solutions in agricultural supply chains?

Agricultural supply chains can benefit from the integration of self-sovereign identity (SSI) solutions, but integration is not without its challenges. These advantages and challenges combine to affect how well SSI can accomplish its mission of making the agricultural supply chain transparent and reliable.

### 4.5.1 Strengths of SSI in agricultural supply chains

#### 4.5.1.1 Enhanced data privacy and security

- **Strength:** Solutions for Self-Sovereign Identity (SSI) emphasize user control over their digital identity and transactional data. It is vital for farmers, distributors, and consumers. SSI's distinguished feature is based on this user-centric design, which empowers end users (farmers) and all their stakeholders to share what they choose and not share what they prefer to keep private.
- **Impact on Agriculture:** This means more accurate, reliable data and a more efficient agricultural supply chain.

#### 4.5.1.2 Decentralized trust model

- **Strength:** SSI solutions utilize decentralized blockchain networks and eliminate the necessity of centralized identity

authorities. This decentralized model enables a trustless environment where you can interact with stakeholders and verify credentials without relying on a single authority.

- **Impact on Agriculture:** In farming, where supply chains often include many participants spread out over a large geography, decentralization means there's less reliance on intermediaries and more direct, safe communication. In agriculture, a secure communications platform that all the diverse, independent partners in a supply chain can trust is absolutely vital. This reduces opportunities for fraud and improves data integrity across cross-border trade (Khoualdi et al., 2025).

#### 4.5.1.3 Improved transparency and traceability

- **Strength:** Blockchain's immutable ledger underpins SSI solutions. This distributed and transparent database records all interactions among the system's stakeholders, allowing them to retrace each supply chain step. From the event that triggers a supply chain notification to the product's arrival at the consumer, stakeholders can see and verify that event.
- **Impact on Agriculture:** Visibility is vital in farming supply chains to guarantee the sourcing and sustainability of food products. SSI can help track a journey from the agricultural field to the final consumers while simultaneously verifying who performed each action through verifiable credentials (Duarte et al., 2024).

#### 4.5.1.4 Compliance with regulatory guidelines

- **Strength:** Storing verifiable credentials allows for compliance with various standards and regulations. For instance, within agricultural supply chains, SSD can support organic certifications, food safety standards, and labor compliance (Shehu and Schneider, 2025).
- **Impact on Agriculture:** This feature helps speed up the process of verifying certifications and regulatory compliance, cutting the time and costs associated with audits and ensuring consistency across international certification bodies.

#### 4.5.1.5 Empowered stakeholders with data ownership and control

- **Strength:** SSI solutions empower stakeholders to control their identity and credential data. Data in control means decisions about what to share and not to share.
- **Impact on Agriculture:** This empowerment results in independence for the individual farmers, who are the main stakeholders in this system and avoid over-exposure of proprietary farming information.

### 4.5.2 Challenges of SSI in agricultural supply chains

#### 4.5.2.1 Technical complexity and accessibility

- **Challenge:** Implementing and managing SSI solutions, especially those based on blockchain, requires a certain level of technical know-how and infrastructure.
- **Impact on Agriculture:** A variety of agricultural parties, particularly small-scale farmers in rural locales, likely do not possess the technical know-how or do not have the luxury of resource abundance needed to adopt SSI systems which may result in unequal adoption rates across regions.

#### 4.5.2.2 Scalability concerns

- **Challenge:** SSI solutions can run into trouble achieving scalability, particularly when they are deployed on blockchain platforms that are not well-suited to handle a large number of transactions or users efficiently.
- **Impact on Agriculture:** The agricultural supply chain consists of many transactions among global networks, which test the sheer ability of SSI platforms to scale. Any delay or inefficiency in processing a transaction will reverberate through the supply chain and impact the user experience affecting stakeholder trust in digital identity systems.

#### 4.5.2.3 Regulatory and legal uncertainties

- **Challenge:** The regulatory environment for SSI and blockchain technology is still evolving, with different standards and regulations across countries.
- **The legal framework for SSI implementation varies significantly across jurisdictions, creating compliance challenges for global agricultural supply chains.** For example, the GDPR in Europe requires clear data controller identification, which can be ambiguous in decentralized systems. This regulatory fragmentation creates significant barriers to widespread adoption, particularly for cross-border agricultural trade that requires consistent identity verification standards.
- **Impact on Agriculture:** Inconsistent global regulatory conditions could hamper seamless cross-border interactions and data sharing in the supply chain. It is a "sticking point" for the agriculture sector if and when it sought to adopt SSI solutions. This is particularly challenging for export-oriented agricultural economies.

#### 4.5.2.4 Interoperability with existing systems

- **Challenge:** Merging self-sovereign identity (SSI) solutions with current identity management, enterprise resource planning (ERP), and supply chain management systems can require across-the-board standardization and interoperability among various platforms (Yildiz et al., 2023).
- **Impact on Agriculture:** Several agricultural supply chains depend on legacy systems for data management. Without interoperability, SSI adoption risks creating parallel systems rather than integrated supply chain infrastructures.

#### 4.5.2.5 Data sovereignty and control

- **Challenge:** The user should definitely own their data, but in an environment like a supply chain, where you have many collaborators, achieving some balance between control and the need to share information gets complex.
- **Impact on Agriculture:** Agricultural supply chains necessitate collaborative data sharing among diverse participants. Ensure that each participant can maintain control over their data to be sent to other parties, which guarantees privacy and transparency but remains a difficult policy and technical issue.

#### 4.5.2.6 Cost and resource requirements

- **Challenge:** Implementing SSI solutions necessitates initial capital expenditure on blockchain infrastructure, secure cryptography, and continuous maintenance.

- **Impact on Agriculture:** These costs may be beyond the reach of many agricultural stakeholders, especially farmers at the small-scale and cooperative levels unless subsidized or supported by governmental programs.

#### 4.5.2.7 User experience and adoption barriers

- **Challenge:** SSI systems require users to manage their online personas, which can entail elaborate procedures for identity authentication and data oversight (Clark and Ruhwanya, 2025).
- **Impact on Agriculture:** Numerous stakeholders in agriculture could face obstacles to using SSI in the near future. This is because many of these stakeholders already struggle to use the digital tools at their disposal, and a poor user experience could slow or discourage adoption.

SSI solutions contribute significantly to transparency, decentralized trust, and stakeholder empowerment within agricultural supply chains. Nonetheless, SSI implementation faces notable challenges, including limited scalability, technological complexity, and compliance with evolving regulatory frameworks. Although SSI provides important benefits—particularly transparency, privacy preservation, and trustless identity verification—the realization of these advantages depends on overcoming barriers related to usability, interoperability, regulatory consistency, and infrastructural maturity. These combined strengths and limitations ultimately determine the practicality and sustainability of SSI adoption in agricultural environments.

## 5 Conclusion

This paper underscores the limitations of existing identity solutions and the potential advantages of SSI in enhancing transparency, security, and privacy within blockchain-enabled agricultural supply chains. Through a systematic literature review (SLR) guided by five research questions, this study explores the role of SSI in overcoming noncompliance with identity principles, establishes criteria for evaluating SSI solutions, and identifies the most suitable SSI solution for agricultural supply chain contexts. From an initial pool of 1,229 articles across four databases, 78 primary articles were selected according to defined criteria, providing a comprehensive review of SSI principles and evaluation criteria specific to the agricultural sector. It is important to note that this assessment of Sovrin's suitability is based on the reviewed literature and theoretical evaluation, and empirical pilot studies are still needed to validate its performance in real agricultural supply chain environments. This study thoroughly reviews key SSI solutions against core principles—user control, privacy, decentralization, and transparency—to determine their fit for agricultural supply chains. Our analysis highlights the strengths, limitations, and operational aspects of each SSI solution. The findings reveal that while no existing SSI solution fully complies with all SSI principles, Sovrin emerges as the most suitable solution, excelling in decentralization, privacy, and traceability. It is important to note that this assessment of Sovrin's suitability is based on the reviewed literature and

theoretical evaluation, and empirical pilot studies are still needed to validate its performance in real agricultural supply chain environments. However, this suitability is comparative rather than absolute, and Sovrin's effectiveness may vary depending on infrastructure maturity, stakeholder readiness, and regional regulatory conditions. It may face limitations in scalability and usability across diverse agricultural contexts. The article further provides promising directions for future research, particularly in advancing secure, user-friendly, and efficient SSI solutions that meet the unique demands of agricultural supply chains. Future work will need to bridge the gap between theoretical evaluations and real-world deployments to validate how SSI performs under operational agricultural constraints.

## 6 Future research

These are some suggested future research directions based on the findings and limitations identified in this paper:

- **Real-World Implementation and Case Studies:** While this paper provides a theoretical evaluation of SSI solutions, real-world case studies that test their performance in active agricultural supply chain environments are needed. Future research could focus on pilot implementations to examine SSI's impact on traceability, security, and stakeholder trust across different agricultural sectors, thereby validating the theoretical assumptions presented in this review.
- **Integration of IoT with SSI in Agricultural Supply Chains:** IoT devices are increasingly used in agriculture to monitor crop and livestock data. Future studies could explore integrating IoT with SSI systems to enable secure, decentralized data sharing from IoT devices, thereby enabling a more data-rich, transparent, and secure agricultural supply chain while also addressing challenges related to device identity, data authenticity, and real-time verification.
- **Enhanced Usability for Non-Technical Stakeholders:** Usability remains a challenge in SSI solutions, particularly for stakeholders in agriculture who may have limited technological expertise. Future studies should explore user-centered design approaches to make SSI solutions more accessible and intuitive for farmers, distributors, and other non-technical users in the supply chain, thereby reducing adoption barriers and promoting inclusive participation.
- **Scalability and Interoperability of SSI Solutions:** While Sovrin aligns well with key SSI principles, it faces challenges with scalability and integration across systems in agricultural supply chains. Future research could focus on developing scalable SSI frameworks that support high-volume data handling and seamless interoperability across diverse platforms and agricultural networks, while maintaining low computational overhead in resource-limited environments.
- **Privacy-Enhanced SSI Architectures:** Privacy is critical in agricultural supply chains due to the sensitivity of trade data and product provenance. Future research could examine integrating advanced privacy techniques, such as differential privacy or homomorphic encryption, into SSI solutions to further enhance data protection without

compromising transparency or increasing system complexity beyond practical deployment capacity.

- Policy and Regulatory Frameworks: For SSI adoption in agricultural supply chains to succeed, regulatory frameworks are essential. Future research could focus on developing guidelines and regulatory frameworks that enable SSI compliance with industry standards while protecting user rights and ensuring secure data handling.
- Evaluation Metrics for SSI in Agricultural Contexts: As the field of SSI in agriculture continues to evolve, there is a need for standardized evaluation metrics tailored to this context. Future research should focus on developing and refining these metrics to accurately assess the effectiveness of SSI solutions in agricultural supply chains, particularly in areas such as trust, security, scalability, and ease of use.
- Exploring Hybrid Models of Centralized and Decentralized Systems: Considering the unique needs of the agricultural sector, future studies could investigate hybrid identity management models that combine centralized and decentralized elements, balancing efficiency, security, and user autonomy to address diverse stakeholder requirements.
- Non-Fungible Tokens (NFTs): Agricultural supply chains can find a key advantage in self-sovereign identity (SSI) solutions built using non-fungible tokens (NFTs). Verifiable digital identities for all forms of life and matter along the supply chain are achievable with NFTs, which offer a unique, immutable record on the blockchain—what amounts to a “certificate of identity” for everything from farmland to farming families and from produce to those who handle it. Integrating NFTs with smart contracts can automate many essential processes along the supply chain to verify identities, thereby reducing fraud, increasing efficiency, and making the overall supply chain more trustworthy. Future research should therefore examine how NFTs can support verifiable identity, provenance authentication, and automated compliance in agricultural environments, while also assessing the scalability and privacy implications of such models.

## Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

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TA: Conceptualization, Formal Analysis, Investigation, Methodology, Visualization, Writing – original draft. GB: Formal Analysis, Investigation, Methodology, Project administration, Supervision, Writing – review and editing. UB: Formal Analysis, Methodology, Resources, Supervision, Validation, Visualization, Writing – review and editing. MP: Investigation, Validation, Visualization, Writing – review and editing.

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